Lakshya Khanna

FULL-STACK ENGINEER

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01 / PROFILE

Full-stack engineer with seven years of front-end and back-end work. I mostly build the systems that keep other systems running: monitoring, speech-to-text pipelines, and AI tools that people use every day.

Right now I'm contracting with CS Technology for **Ferroglobe (US)**, building monitoring and AI tools for industrial plants where downtime costs real money. I've worked offshore with US teams for most of my career, so different timezones and async communication are normal for me. I like work that is quiet and reliable, like a tool I wrote months ago that is still running and nobody has had to think about.

02 / PROJECTS

AI Log Analyzer & Auto-Debugger

open source

A Python agent that watches multi-service apps and catches crashes as they happen. It reads the stack trace, works out where things went wrong, and suggests a code fix for the front end or back end. The Slack and email alerts include the proposed diff, not just the error message.

Python · OpenAI · FastAPI · Slack API · SMTP

github.com/lakshyak789/AI_log_analyzer

AI Call & Interview Copilot

open source

A free alternative to paid call assistants. It listens to live audio and gives advice during the call, turns a UI screenshot into React code, and reads your résumé to come up with 15 interview questions based on it. Most people found it because the useful parts are free.

Python · React · Whisper · Vision AI · WebRTC

github.com/lakshyak789/AI_meeting-copilot

Chess Battlefield

side project

3D battle chess that runs in the browser. Every piece is a warrior with its own kill animation. The rook is a war elephant that picks up its victim with its trunk and throws it off the board, and on checkmate the attacking pieces walk over and finish off the king in a slow close-up.

- Full chess rules (castling, en passant, promotion, game clocks) with Stockfish compiled to WebAssembly as the computer opponent, at three difficulty levels
- Play a friend online over WebRTC, peer to peer, with no backend server at all
- Rigged 3D models with skeletal animation, four graphics quality levels, and touch controls so it also runs on phones

three.js · WebGL · Stockfish WASM · WebRTC · JavaScript

chess.lakshya.space

CS Technology LLC · Ferroglobe (US)

Apr 2026 - Present

Contract Full-Stack Engineer, working offshore

Plant Health Management System. A monitoring platform for Ferroglobe's US plants, where downtime is expensive and someone needs to know the moment a service goes down.

- Admin panel for managing environments (UAT, Prod) and service config (REST endpoints, databases, cron jobs)
- Python heartbeat agent that checks every service once a minute and feeds a live green/red status dashboard
- Incident tracking, scheduled maintenance windows, and email alerts people can subscribe to per service
- CRUD APIs for service config, plus bug fixes and new features on an older Django and React TypeScript app

React · FastAPI · PostgreSQL · Python · Django

MES Assist. An AI chatbot platform for a Manufacturing Execution System. Admins set up bots from a no-code panel, and the chat widget goes into any page with a single script tag.

- RAG and a data agent built on LlamaIndex, Qdrant and HuggingFace. It answers from documents, runs read-only SQL and API calls, and shows live plant data as tables or one of 11 chart types
- Widget is a Vite IIFE bundle inside a Shadow DOM, so the host page's CSS can't break it
- Each chatbot can use a different LLM provider (Azure OpenAI, OpenRouter or OpenAI), and the model, temperature, prompt and RAG settings are all editable from the UI
- The widget forwards the host app's JWT in a custom header, and the server checks it before returning any plant data
- Nginx serves the admin panel and widget on the same origin as the main MES app, so there are no exposed ports, CORS problems or mixed-content warnings

Django · DRF · React · TypeScript · LlamaIndex · Qdrant · Docker · nginx · Azure OpenAI

Isha Foundation · Whisper AI pipelines

Jul 2024 - Dec 2025

Full-Stack Engineer, speech-to-text

Built speech-to-text pipelines on Whisper AI for an internal transcription service, working on both the FastAPI back end and the React front end.

- Added Tamil transcription with Whisper.cpp. A FastAPI router splits audio into 15-second chunks with FFmpeg and stitches the text back together
- Fine-tuned Whisper on over 400 hours of audio on Kaggle, working around GPU and memory limits
- Added threading and a task queue to the Flask service so jobs take turns on the GPU instead of slowing each other down
- Wrote a Python tool that takes audio, transcribes it, posts it to Confluence and opens a Jira ticket
- Built a Confluence page diff tool (FastAPI and React) that opens inside the page through Tampermonkey
- Set up a drag-and-drop UI on Dufs for batch audio jobs that produce .srt subtitle files

Whisper.cpp · FastAPI · React · FFmpeg · Kaggle

Isha Foundation · Sadhanapada Volunteer

Jun 2023 - May 2024

Web team: React, Apps Script, Looker Studio

Worked on the organisation's website in React, then moved on to building reports with Google Apps Script and Looker Studio.

Ibexlabs · Labra.io (US)

Apr 2021 - Oct 2022

Software Engineer, working offshore

Worked on Labra.io (US), an AWS Marketplace SaaS product. I owned the ConnectWise CRM integration from start to finish and did full-stack work across the rest of the product.

- Front end in TypeScript, React and Redux, tested with Jest
- Back-end APIs in Go and Python on PostgreSQL

TypeScript · React · Redux · Go · Python · PostgreSQL

SupportGenie
Mar 2019 - Mar 2021

React.js Developer

I was the only front-end developer on a ticketing portal. Moved the marketing site from AngularJS to Gatsby, replaced REST calls with Apollo GraphQL, and built a WebRTC video chat app from scratch.

- Real-time ticket updates with Autobahn pub/sub
- Set up Jenkins CI and moved services into Docker containers
- UI built with Material-UI on top of Python APIs

Actionable Science
Nov 2018 - Mar 2019

JavaScript Developer (Intern)

My first job. Built the React UI for a chatbot platform where the conversation flows were written in YAML. It taught me to read other people's config files very carefully.

Axel Web Technology
Jan 2016 - Apr 2016

Backend PHP Developer

Worked on the back end of an electronic health records system in CakePHP. This is where I first found out what legacy code really means.

04 / SKILLS

Front end	React · TypeScript · Redux · Next.js · three.js · Gatsby · Apollo GraphQL · Material UI · Jest
Back end	Python · FastAPI · Django · Node · Go · PostgreSQL · REST · GraphQL
AI & audio	Whisper · Whisper.cpp · FFmpeg · fine-tuning on Kaggle · RAG (LlamaIndex, Qdrant) · Vision AI
DevOps & tools	AWS · Docker · nginx · Linux · Jenkins · WebRTC · Autobahn pub/sub · Slack/email automation · Confluence/Jira scripting

05 / EDUCATION

Chandigarh Engineering College
2016 - 2018

Master's in Computer Applications

Lovely Professional University
2012 - 2015

Bachelor's in Computer Applications

Find me at: linkedin.com/in/lakshya-khanna-8b861754 · github.com/lakshyak789 · https://www.lakshya.space